Biscale Convolutional Self-Attention Network for Hyperspectral Coastal Wetlands Classification

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***Abstract*— Coastal wetlands classification is a hot but challenging issue. Hyperspectral image (HSI) can provide abundant spectral information for coastal wetlands, and deep learning excels at extracting abstract features. However, effectively leveraging global and local features to enhance the accuracy of coastal wetlands classification remains a significant challenge. In this letter, we propose a biscale convolutional self-attention network (termed as HyperBCS) for hyperspectral coastal wetlands classification. HyperBCS consists of biscale adding module (BSAM) and convolutional self-attention module (CSM). On the one hand, BSAM uses two branches to extract features of different scales. On the other hand, CSM is a paralleled structure of convolution and self-attention, which can effectively extract both local and global features. Experiments on two hyperspectral datasets captured by Zhuhai-1 satellite indicate that HyperBCS can improve the accuracy of hyperspectral coastal wetlands classification, showcasing the highest accuracy (OA = 98.29% and 96.82%, Kappa = 0.976 and 0.958 in two datasets) compared with other methods. Our code and datasets are available at https://github.com/JeasunLok/HyperBCS.**

***Index Terms*— Coastal wetlands classification, convolution, deep learning, hyperspectral image (HSI), self-attention mechanism.**

I. Introduction

COASTAL wetlands play a crucial role in biological, ecological, and hydrological processes [1]. Classification is the most fundamental aspect in the field of wetland science. Hyperspectral image (HSI), which contains hundreds of narrow contiguous spectral bands and rich spectral information, has great potential for coastal wetlands classification.

Traditional hyperspectral wetlands classification methods typically rely on machine learning techniques, which include decision tree [2], support vector machine [3], random forest [4], and artificial neural networks [5]. However, constrained to unsatisfactory feature extraction ability in remote sensing data, traditional machine learning methods struggle with complex coastal wetlands classification tasks [6]. In recent years, deep learning techniques, especially convolutional neural networks (CNNs) with a strong feature extraction ability, has obtained excellent performances in coastal wetlands classification [7], [8], [9], [10]. While proficient in local feature extraction, CNN's limitations in global feature extraction due to limited receptive field hinder its effectiveness in classifying coastal wetlands with high confusion and variability [11].

Faced with the above-mentioned challenges, transformer with self-attention mechanism is proposed [12] and has been applied to hyperspectral classification [13], providing us a new perspective in global features extraction. For instance, Liu et al. [14] and Gao et al. [15] both design transformer structures applicable to HSI to classify coastal wetlands. Moreover, the combined approach involving CNN and transformer is employed to enhance the accuracy of hyperspectral classification [16], with coastal wetlands classification following suit [17], [18]. Despite remarkable performances, sequential structures of convolution and self-attention mechanism in these methods cause more feature loss compared with paralleled structure [19]. For example, in a sequential setup, starting with convolution leads to global feature loss and the subsequent self-attention mechanism operates on this basis, which has already suffered irreversible feature loss.

In this letter, we propose a biscale convolutional self-attention network (termed as HyperBCS), which consists of biscale adding module (BSAM) and convolutional self-attention module (CSM), for coastal wetlands classification. The main innovative contributions are twofold: 1) the two-branch structure can extract features of two different scales without significantly increasing parameters. As a plug-and-play module, it can be incorporated into any hyperspectral coastal wetlands classification models, yielding an enhanced performance in a simple way and 2) a novel paralleled structure of convolution and self-attention mechanism, regarded as a hybrid operation module which considers the convolution and self-attention mechanism as having equal status and extracting features from the same feature maps meanwhile, is proposed, which pioneers a new line of thought for hyperspectral coastal wetlands classification. To the best of our knowledge, the paralleled structure proposed in this letter is the first attempt in hyperspectral coastal wetlands classification. Moreover, it should be noted that our proposed HyperBCS has versions with 1-D structure (1D-HyperBCS), 2-D structure (2D-HyperBCS), and 3-D structure (3D-HyperBCS),

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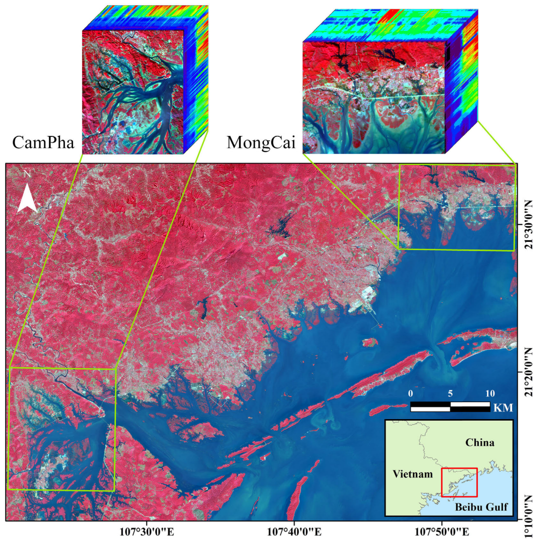

Fig. 1. Study area and HSI datasets (i.e., MongCai and CamPha) acquired by the Zhuhai-1 satellite with sensor OHS-1 in the study.

with the only difference being the dimension of the convolution.

II. Study Area and Datasets

The study area is located in Quang Ninh Province, Vietnam, the west of Beibu Gulf. Two datasets are used in this study (see Fig. 1). The first dataset MongCai is located in the west of Móng Cái (107°46′E–107°55′E, 21°28′N-21°34′N) and was taken on January 19, 2022, while the second dataset CamPha is located in Câm Phà (107°20′E-107°28′E, 21°12′N–21°20′N) and was taken on December 22, 2019. Both of them were acquired by the Zhuhai-1 satellite with sensor OHS-1. The spatial resolution is 10 m, spectrum resolution is 2.5 nm and the spectral range is from 400 to 1000 nm.

The coastal wetlands classification system adopted in this study include river, lake, coastal swamp, coastal marsh, salt pans, tidal flat, aquaculture, and reservoir, which is suited for the coastal wetlands in Southeast Asia. In addition, forest, impervious, brick house, and bare are also added into our classification system as nonwetland classes. The ground truth information of our datasets are annotated by ecological experts of research teams and validated by very high-resolution satellite images and field research.

III. Proposed Method

A. Overall Architecture

The overall architecture of the proposed 3D-HyperBCS is shown in Fig. 2. The input HSI patch is first processed by BSAM, which is composed of two branches. The first branch adopts the original patch as input while the second branch takes half of the original patch as input. The results of both branches are then added together and fed into CSM, which aims at capturing local features by convolution and global features by self-attention mechanism in a paralleled structure. Finally, the global average pooling and fully connected layer are used to obtain the classification results.

B. BiScale Adding Mechanism

BSAM takes the input patch $\boldsymbol{M_P}$ with size $P \times P \times C$ and generates the feature $\boldsymbol{M'}$. There are two branches in BSAM. The first branch takes $\boldsymbol{M_P}$ as input. The second branch extracts the central patch $\boldsymbol{M_p}$ with size $(P/2) \times (P/2) \times C$ as input. The output feature $\boldsymbol{M'}$ is obtained by

$$\boldsymbol{M'} = \mathcal{F}_P(\boldsymbol{M_P}) + \mathcal{F}_p(\boldsymbol{M_p}) \tag{1}$$

where $\mathcal{F}_P$ denotes the global feature extraction module and $\mathcal{F}_p$ denotes the local feature extraction module.

C. Convolutional Self-Attention Module

CSM is a paralleled feature extraction module of convolution and self-attention, in which the $\boldsymbol{M'}$ is successively processed by a convolution layer, a batch normalization (BN) layer, and a ReLU layer, and then the feature cube $\boldsymbol{M}$ with size $B \times C_{\text{in}} \times D \times P \times P$ is generated. Afterward, $\boldsymbol{M}$ is fed into the CS Mechanism (see Fig. 3). Three $1 \times 1 \times 1$ 3D convolution layers are carried out to obtain $\boldsymbol{K}_i$, $\boldsymbol{Q}_i$ and $\boldsymbol{V}_i$, respectively, which will be applied to the self-attention part and convolution part at the same time

$$\boldsymbol{K}_i = \mathrm{C}_{1,K}(\boldsymbol{M}) = W_K^{(i)}\boldsymbol{M} \tag{2}$$

$$\boldsymbol{Q}_i = \mathrm{C}_{1,Q}(\boldsymbol{M}) = W_Q^{(i)}\boldsymbol{M} \tag{3}$$

$$\boldsymbol{V}_i = \mathrm{C}_{1,V}(\boldsymbol{M}) = W_V^{(i)}\boldsymbol{M} \tag{4}$$

where $\mathrm{C}_{1,K}$, $\mathrm{C}_{1,Q}$ and $\mathrm{C}_{1,V}$ denote the $1 \times 1 \times 1$ 3-D convolution layers, respectively, $W_K^{(i)}$, $W_Q^{(i)}$ and $W_V^{(i)}$ indicate parameters of the ith self-attention head.

For the self-attention part, the unfold operation with specific kernel AttKernel is performed to extract the multihead attention

$$\mathbf{Att}(\boldsymbol{M}) = \bigsqcup_i \left\{ \mathrm{softmax}\left[\frac{\left(W_Q^{(i)}\boldsymbol{M}\right)\left(W_K^{(i)}\boldsymbol{M} + \zeta\right)^T}{\sqrt{d}} \right] W_V^{(i)}\boldsymbol{M} \right\} \tag{5}$$

where $\sqcup_i$ denotes concatenation operator of i vectors, ζ denotes the positional embedding, and d denotes the dimension of $\boldsymbol{K}_i$, $\boldsymbol{Q}_i$, and $\boldsymbol{V}_i$.

For the convolution part, $\boldsymbol{K}_i$, $\boldsymbol{Q}_i$, and $\boldsymbol{V}_i$ are concatenated and reshaped into a single feature cube, which is then successively processed by a $1 \times 1 \times 1$ convolution layer and a $3 \times 3 \times 3$ convolution layer. The result of the convolution part is obtained by

$$\mathbf{Conv}(\boldsymbol{M}) = \mathrm{C}_3\left\{ \mathrm{C}_1\left[\bigsqcup_i \left(W_K^{(i)}\boldsymbol{M}, W_Q^{(i)}\boldsymbol{M}, W_V^{(i)}\boldsymbol{M} \right) \right] \right\} \tag{6}$$

where $\sqcup_i$ denotes concatenate i vectors, C_1 denotes $1 \times 1 \times 1$ 3-D convolution, and C_3 denotes $3 \times 3 \times 3$ 3-D convolution.

Finally, the self-attention part and the convolution part are summed with learnable weights, and the output of CSM is obtained by

$$\mathbf{CSM}(\boldsymbol{M}) = \alpha \cdot \mathbf{Att}(\boldsymbol{M}) + \beta \cdot \mathbf{Conv}(\boldsymbol{M}) \tag{7}$$

where α and β denote the learnable weight of the self-attention part and convolution part, respectively.

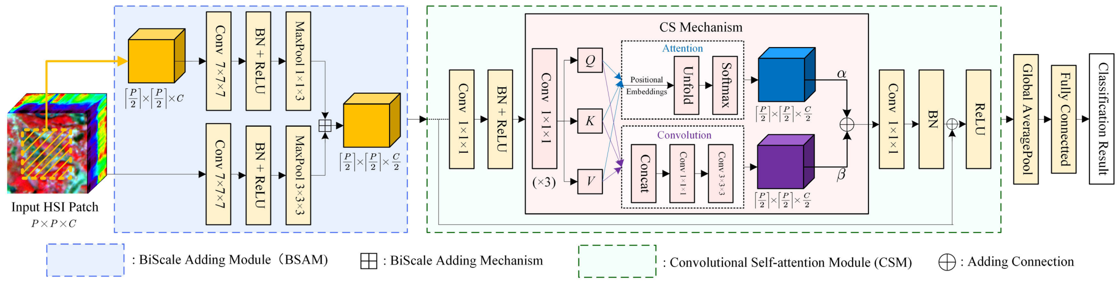

Fig. 2. Overall architecture of 3D-HyperBCS.The input HSI patch is first processed by BSAM. After that, the feature cubes will go through the CSM to integrate the features extracted by 3-D-convolution and self-attention mechanism in a paralleled way. Eventually, the classification result can be obtained by a global average pooling layer and a fully connected layer successively.

TABLE I
CLASSIFICATION ACCURACIES FOR DATASET MONGCAI WHEN USING 200 TRAINING SAMPLES

Class	Training	Testing	2D-CNN	3D-CNN	3D+1D-CNN	HybridSN	1D-RNN	ViT	SpectralFormer	Swin transformer	1D-HyperBCS	2D-HyperBCS	3D-HyperBCS
Forest	200	106481	**99.97**	97.53	99.31	99.79	95.22	93.63	91.32	99.92	98.10	99.94	99.52
Impervious	200	2802	95.72	93.36	95.75	97.32	96.04	92.22	93.47	**99.75**	97.39	97.86	99.04
Bare	200	2025	**100**	96.54	98.91	98.86	88.15	87.31	95.26	98.46	96.89	95.65	98.96
River	200	17360	95.57	92.71	94.08	95.39	89.48	81.49	96.34	91.81	94.77	90.26	**97.96**
Lake	200	5989	98.18	97.96	99.23	99.05	96.78	91.50	**99.33**	98.75	97.91	97.98	99.25
Coastal swamp	200	2623	96.19	92.91	92.07	**97.94**	91.80	88.30	93.18	97.14	91.92	92.79	94.24
Coastal marsh	200	99611	94.34	84.79	88.93	89.13	83.67	81.82	84.69	92.53	88.22	95.15	**96.65**
Salt pans	200	19724	97.32	97.82	98.57	99.23	98.78	97.42	99.89	98.97	99.11	98.96	**99.90**
Tidal flat	200	10393	97.33	94.33	96.36	98.67	89.94	87.15	93.37	98.48	95.06	97.17	**98.69**
Aquaculture	200	598	99.67	89.97	95.32	99.67	87.96	95.48	95.15	**99.83**	93.81	91.47	99.16
Reservoir	200	3483	49.33	95.00	97.56	97.35	89.81	80.36	91.47	98.21	98.82	98.19	**98.85**
OA(%)			96.55	92.30	94.85	95.42	90.54	88.21	90.16	96.48	94.10	97.21	**98.29**
Kappa			0.951	0.893	0.928	0.936	0.870	0.838	0.864	0.950	0.918	0.961	**0.976**
Params(M)			0.104	**0.002**	0.033	1.472	0.042	0.092	0.098	3.451	0.125	0.259	0.113
FLOPs(M)			1.046	**0.078**	1.329	146.280	0.918	3.013	3.431	77.680	3.984	18.007	149.473

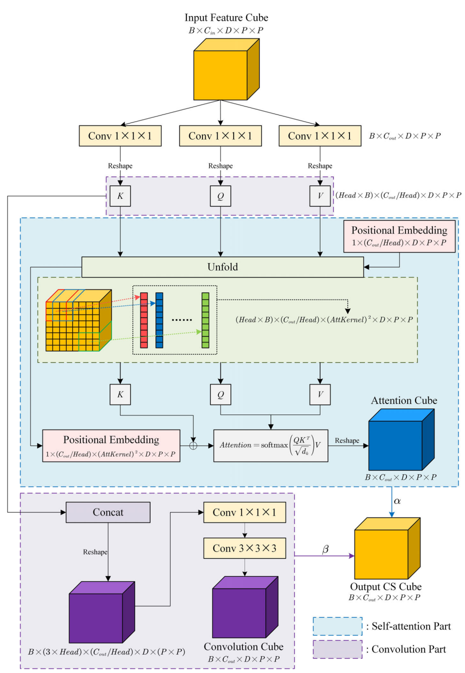

Fig. 3. CS mechanism in CSM of 3D-HyperBCS. Three 1 × 1 × 1 3-D convolution layers are carried out to obtain key feature cubes, then three key feature cubes are simultaneously fed into convolution part and self-attention part to extract in-depth and hidden features in a paralleled way, respectively. At last, the self-attention part and the convolution part are summed with learnable weights to get the output CS cube.

IV. EXPRIMENTS AND RESULTS

A. Implementation Details

To validate the effectiveness of the proposed methods, we compare the proposed HyperBCS with eight state-of-the-art HSI classification methods, including CNN based models such as 2D-CNN [7], 3D-CNN [8], 3D + 1D-CNN [9], HybridSN [20], recurrent neural networks with 1-D structure (1D-RNN) [10], and self-attention mechanism-based methods including ViT [21], SpectralFormer [13], and Swin transformer [22]. It should be noted that all of the aforementioned methods are reproduced as closely as possible to original references. Therefore, CNN-based methods with different dimensions are not simply substituting convolutions and their structures are quite different from one another.

In the training stage, 200 samples for each class are used as training set, and the rest samples are used as testing set. The Adam algorithm is used to minimize the cross-entropy loss, the learning rate, momentum, weight decay, and patch size are set to 0.0005, 0.9, 0, and 9, respectively.

B. Experimental Results and Comparison

Five indices [i.e., accuracy of each class, overall accuracy (OA), kappa coefficient (Kappa), model parameters (Params), and floating point operations (FLOPs)] are used to validate the classification models. The classification accuracies are shown in Tables I and II, while the visualization results are shown in Figs. 4 and 5. It should be noted that even though the Params(M) and FLOPs(M) of some methods including the new added HybridSN, Swin transformer and our proposed HyperBCS are much higher than other methods, the training and testing time of 100 epoch of all methods are less than half-an-hour with a single GPU (Nvidia GeForce RTX2080ti), which means the computation cost of all methods is acceptable in mostly all situations.

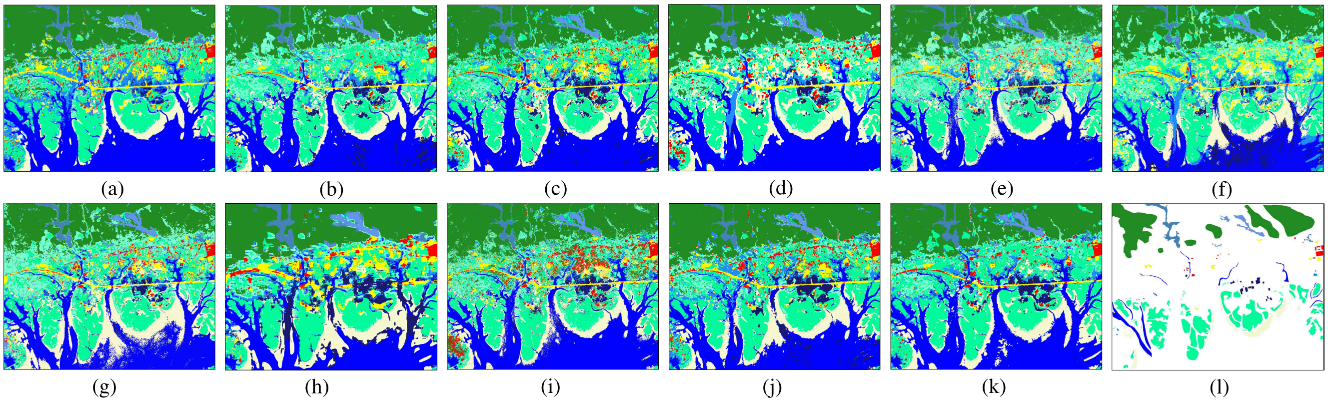

Fig. 4. Classification maps of different methods for the dataset MongCai. The colors in the classification maps represent the same land cover categories as the background colors in Tables I and II. (a) 2D-CNN. (b) 3D-CNN. (c) 3D + 1D-CNN. (d) HybridSN. (e) 1D-RNN. (f) ViT. (g) SpectralFormer. (h) Swin transformer. (i) 1D-HyperBCS. (j) 2D-HyperBCS. (k) 3D-HyperBCS. (l) Ground truth.

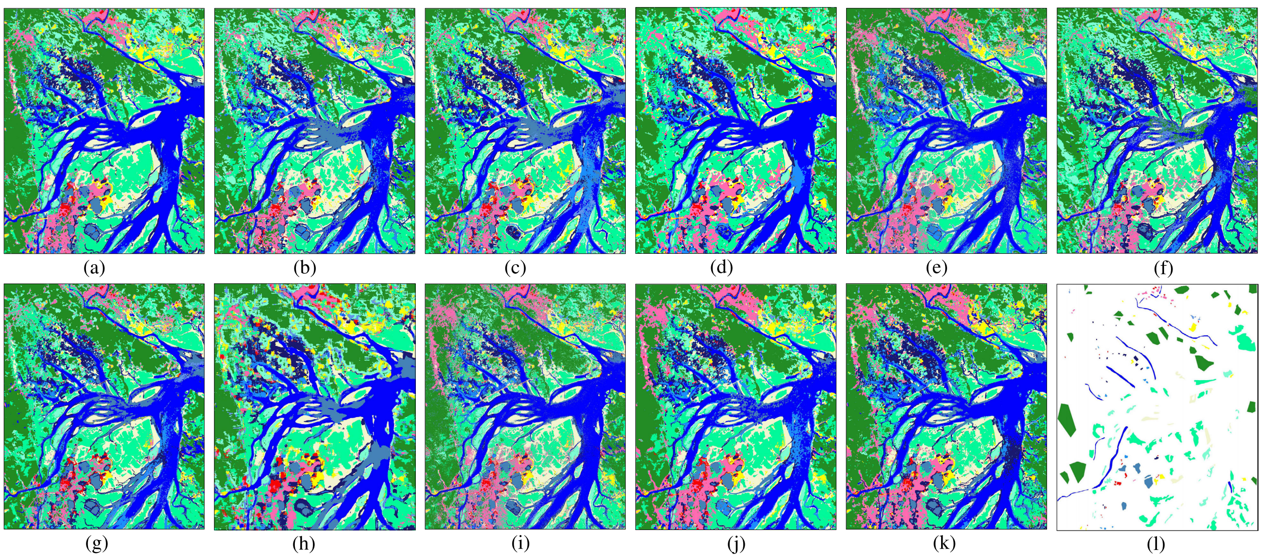

Fig. 5. Classification maps of different methods for the CamPha. The colors in the classification maps represent the same land cover categories as the background colors in Tables I and II. (a) 2D-CNN. (b) 3D-CNN. (c) 3D + 1D-CNN. (d) HybridSN. (e) 1D-RNN. (f) ViT. (g) SpectralFormer. (h) Swin transformer. (i) 1D-HyperBCS. (j) 2D-HyperBCS. (k) 3D-HyperBCS. (l) Ground truth.

TABLE II

CLASSIFICATION ACCURACIES FOR DATASET CAMPHA WHEN USING 200 TRAINING SAMPLES

Class	Training	Testing	2D-CNN	3D-CNN	3D+1D-CNN	HybridSN	1D-RNN	ViT	SpectralFormer	Swin transformer	1D-HyperBCS	2D-HyperBCS	3D-HyperBCS
Forest	200	56070	89.73	80.87	91.59	89.05	89.58	70.11	89.55	93.32	89.30	94.21	**97.73**
Impervious	200	1742	97.36	92.48	91.39	98.74	85.48	89.55	98.00	98.33	96.27	98.56	**98.97**
Brick house	200	1496	99.73	95.45	96.79	**100**	98.13	96.12	93.98	99.60	98.06	**100**	99.93
Bare	200	4981	**98.88**	71.13	96.04	97.79	83.98	81.83	76.85	96.89	86.77	97.71	97.59
River	200	16920	97.03	87.30	93.76	96.22	86.58	80.06	92.50	**96.23**	90.61	95.12	94.52
Lake	200	304	**100**	93.42	**100**	**100**	91.12	96.38	97.70	**100**	99.01	**100**	**100**
Coastal swamp	200	12884	94.16	91.29	89.62	95.86	82.65	92.12	75.11	95.30	93.33	93.33	**96.41**
Coastal marsh	200	57765	97.19	92.52	95.54	93.78	93.63	88.35	94.52	95.63	89.40	**98.67**	95.68
Salt pans	200	9585	**99.97**	95.44	95.44	98.93	95.26	97.51	99.21	98.48	99.32	99.89	99.61
Tidal flat	200	6780	98.45	94.60	97.43	99.05	95.10	92.30	97.43	**99.63**	96.27	98.73	98.42
Aquaculture	200	1154	99.31	80.07	95.49	99.74	74.44	79.38	93.85	99.57	93.85	99.91	**100**
Reservoir	200	1814	98.13	89.97	98.62	**99.61**	77.29	97.24	97.68	99.17	93.50	97.96	**99.61**
OA(%)			94.81	87.64	93.72	93.49	90.30	82.46	91.17	95.48	90.77	96.52	**96.82**
Kappa			0.933	0.840	0.918	0.915	0.874	0.777	0.884	0.941	0.881	0.955	**0.958**
Params(M)			0.104	**0.002**	0.033	1.472	0.042	0.092	0.098	3.451	0.125	0.259	0.113
FLOPs(M)			1.046	**0.078**	1.329	146.280	0.918	3.013	3.431	77.680	3.984	18.007	149.473

Some observations can be highlighted from the above-mentioned classification results. On the one hand, among all methods, the proposed 3D-HyperBCS obtains the best performance. For instance, 3D-HyperBCS achieves the highest OA and Kappa in both datasets. Except for HyperBCS models, 2D-CNN and Swin transformer achieve good results in both datasets and they perform best in some classes. However, 2D-CNN misclassifies half of reservoirs and Swin transformer misclassifies many rivers and coastal marsh in dataset MongCai while HyperBCS models perform more

TABLE III
ABLATION EXPERIMENTS IN TWO DATASETS

Baseline	CSM	BSAM	MongCai		CamPha		Params(M)	FLOPs(M)
			OA(%)	Kappa	OA(%)	Kappa		
1D-HyperBCS	×	–	92.41	0.895	88.36	0.851	**0.095**	**3.025**
	✓	–	**94.10**	**0.918**	**90.77**	**0.881**	0.125	3.984
2D-HyperBCS	×	×	95.93	0.943	93.21	0.912	0.178	15.333
	×	✓	97.04	0.958	95.02	0.935	0.278	18.553
	✓	×	97.05	0.958	95.23	0.938	**0.159**	**14.787**
	✓	✓	**97.21**	**0.961**	**96.52**	**0.955**	0.259	18.007
3D-HyperBCS	×	×	94.88	0.927	94.39	0.927	0.174	167.909
	×	✓	95.14	0.931	94.96	0.934	0.196	190.650
	✓	×	97.34	0.962	95.43	0.940	**0.092**	**126.732**
	✓	✓	**98.29**	**0.976**	**96.82**	**0.958**	0.113	149.473

averagely. Moreover, the classification maps of 3D-HyperBCS [see Figs. 4(k) and 5(k)] are more aggregated and the boundaries are much smoother than other methods, demonstrating the superiority of the paralleled structure. On the other hand, HyperBCS with different dimension perform better than CNN-based models with the same dimension. As can be seen in Tables I and II, 2D-HyperBCS and 3D-HyperBCS perform better than 2D-CNN and 3D-CNN in both datasets, respectively. This is because the paralleled structure of CNN and self-attention mechanism in HyperBCS methods incorporates novel features extracted by the self-attention mechanism compared with CNN based methods.

C. Ablation Study

Ablation studies are performed to further verify the effectiveness of the proposed HyperBCS. When BSAM is not used, HyperBCS models only utilize $\boldsymbol{M_P}$ as input to extract feature in single scale. When CSM is not used, HyperBCS models use convolution with same dimension instead. It can be found from Table III that when BSAM or CSM is used, OA and Kappa of HyperBCS with different dimension are all improved. The accuracy can be further improved when both BSAM and CSM are used in 2D-HyperBCS and 3D-HyperBCS. Furthermore, BSAM improves the performance in a very simple way, and the parameters of CSM are less than 2-D and 3-D convolutions.

V. CONCLUSION

In this letter, we have proposed a novel deep learning-based method, namely HyperBCS, for hyperspectral coastal wetlands classification. The BSAM and CSM in the HyperBCS are used to obtain higher accuracy without significantly increasing parameters. BSAM extracts features in different scales with two branches, whereas the underlying idea of CSM is to extract features with convolution and self-attention mechanism in a paralleled way. The effectiveness of the proposed HyperBCS are verified by two hyperspectral datasets (i.e., MongCai and CamPha) from Zhuhai-1 satellite with OHS-1 sensor. Future research could be devoted, for instance, to applying HyperBCS to other hyperspectral coastal wetlands datasets and further exploring its potential in other hyperspectral classification tasks.

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REFERENCES

[1] B. Hosseiny, M. Mahdianpari, B. Brisco, F. Mohammadimanesh, and B. Salehi, "WetNet: A spatial–temporal ensemble deep learning model for wetland classification using Sentinel-1 and Sentinel-2," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4406014.

[2] K. Liu, X. Li, X. Shi, and S. Wang, "Monitoring mangrove forest changes using remote sensing and GIS data with decision-tree learning," *Wetlands*, vol. 28, no. 2, pp. 336–346, Jun. 2008.

[3] C. Doña et al., "Monitoring hydrological patterns of temporary lakes using remote sensing and machine learning models: Case study of la Mancha Húmeda biosphere reserve in central Spain," *Remote Sens.*, vol. 8, no. 8, p. 618, Jul. 2016.

[4] J. Cabezas, M. Galleguillos, and J. F. Perez-Quezada, "Predicting vascular plant richness in a heterogeneous wetland using spectral and textural features and a random forest algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 646–650, May 2016.

[5] N. E. Mitrakis, C. A. Topaloglou, T. K. Alexandridis, J. B. Theocharis, and G. C. Zalidis, "Decision fusion of GA self-organizing neuro-fuzzy multilayered classifiers for land cover classification using textural and spectral features," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2137–2152, Jul. 2008.

[6] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[7] V. Sharma, A. Diba, T. Tuytelaars, and L. Van Gool, "Hyperspectral CNN for image classification & band selection, with application to face recognition," ESAT, Leuven, KU Leuven, Belgium, Tech. Rep., KUL/ESAT/PSI/1604, 2016.

[8] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[9] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[10] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[11] B. Cui, X. Li, J. Wu, G. Ren, and Y. Lu, "Tiny-scene embedding network for coastal wetland mapping using Zhuhai-1 hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[12] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[13] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[14] K. Liu et al., "Mapping coastal wetlands using transformer in transformer deep network on China ZY1–02D hyperspectral satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3891–3903, 2022.

[15] Y. Gao et al., "Fusion classification of HSI and MSI using a spatial–spectral vision transformer for wetland biodiversity estimation," *Remote Sens.*, vol. 14, no. 4, p. 850, Feb. 2022.

[16] S. Hao, Y. Xia, L. Zhou, Y. Ye, and W. Wang, "Spectral and spatial feature fusion for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[17] Z. Xie, J. Hu, X. Kang, P. Duan, and S. Li, "Multilayer global spectral–spatial attention network for wetland hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518913.

[18] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.

[19] X. Pan et al., "On the integration of self-attention and convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 805–815.

[20] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[21] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2020, pp. 1–22.

[22] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.